Flip class 1 assignment: Cloud Computing

1. Identify the limitation of traditional Computing faced by ShopEase.

- 1. Limited Scalability to handle 5-6x traffic during festive seasons.
  2. Frequent downtime due to Server Overload.
  3. Hardware overheating in on-premis infrastructure.
  4. High maintenance and repair cost.
  5. Large upfront investment in Server and networking equipment.
  6. Slow resource provisioning and manual upgrads.
  7. Limited disaster recovery and backup system.

2. Explain how cloud Computing Can Solve the issue of Seasonal traffic spikes.

- 1. provids auto-Scaling to increase / decrease resources automatically.
  2. offer on-demand resource allocation during peak traffic.
  3. uses load balancing to distributed traffic efficiently.
  4. ensure high availability across multiple data Centers.
  5. Eliminats hardware Overheating issues
  6. pay-as-you-use model reduces unnecessary

cost.

7. Faster deployment of additional server within minutes.

3. Compare Capital expenditure (CapEx) in traditional Computing with operational expenditure (OpEx) in cloud Computing for this Case.

— A) Traditional Computing (CapEx)

1. High upfront cost for purchasing hardware.
2. Investment in Cooling, power and office space.
3. Cost of IT Staff for maintenance.
4. Hardware replacement and upgrade expenses.
5. payment required even during low usage.
6. Long-term financial Commitment
7. Depreciation of physical assets.

B) Cloud Computing (OpEx)

1. No upfront hardware investment
2. pay-as-you use pricing model
3. Reduced maintenance responsibility.
4. No infrastructure management cost.
5. Cost increase only during high usage (e.g. Diwali)

6. Flexible monthly or usage-based billing.
7. No hardware depreciation cost

4. Suggest an appropriate cloud deployment model (public / private / Hybrid) with justification.

- Suggested model : public cloud (such as Amazon web Services or Microsoft Azure)
  - i. Cost-effective for startup.
  - ii. Highly scalable for seasonal traffic spiks.
  - iii. No need to manage physical infrastructure.
  - iv. Built-in security and Compliance features.
  - v. Easy and fast deployment.
  - vi. Global availability and reliability
  - vii. Suitable for a e-commerce workloads.

5. What risks should the Company Consider before migrating to the cloud?

→ 1. Data security and privacy Concerns.
2. vendor lock-in with one Cloud provider
3. Possible Cloud Service outage.
4. Data migration Complexity and cost
5. Compliance with data protection regulation
6. Dependency on internet Connectivity
7. Risk of unexpected or hidden Costs.